Multiple IV Bolus Practice 3

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

A 80 kg patient is scheduled to receive a 20 mg/kg IV bolus injection of a medication every 6 hours. The drug has an apparent volume of distribution that is 25% of body weight and a total body clearance is 4.62 L/hr. Determine the following for this medication in this patient:

a Elimination half-life.

b Maximum plasma drug concentration of the first dose.

c Minimum plasma drug concentration of the first dose.

d Maximum steady-state plasma drug concentration.

e Minimum steady-state plasma drug concentration.

f Average steady-state plasma drug concentration.

g Average amount of drug in the body during steady state.

h Plasma level 12 hours after the last dose, assuming steady-state levels are achieved.